

Distributions in Data Science – Notes

1. What is Distribution?

- Distribution in data science is a method which shows the probable values for a variable and how often they occur. It is a simple way to visualize a set of data..
- It shows:
 - Possible values of a variable
 - How often each value occurs (probabilities)

Example:

- Coin toss:
 - Head = 0.5
 - Tail = 0.5
 - Any other outcome = 0

Important Points

- Distribution includes **all possible outcomes**.
- It is often shown using **graphs**, but:
 - Graph is only a **visual representation**
 - Actual distribution is defined by **probabilities**

Example of Distribution

Tossing Two Coins

Possible outcomes:

HH, HT, TH, TT

Probability:

Each = 0.25

This is an example of Uniform Distribution
(all outcomes equally likely)

Types of Data in Distribution

The data can be classified into two types

1. Discrete Data
2. Continuous Data

(A) Discrete Data

- Takes **fixed values only**
- Example:
 - Pass / Fail
 - Number of students

(B) Continuous Data

- Can take **any value in a range**
- Example:
 - Height
 - Weight
 - Temperature

So based on type of data, distribution can be of two types.

Discrete probability distribution

Continuous probability distribution

Poisson Distribution

- Used to find probability of events happening in a fixed interval
- Example: Number of calls received in a call center per hour

Binomial Distribution

- Used when there are only two outcomes (Success/Failure)
- Example: Tossing a coin multiple times

Bernoulli distribution

It is the **simplest probability distribution** in data science.

It deals with **only one trial** having **two possible outcomes**:

- Success (1)
- Failure (0)

Examples:

- Toss a coin → Head (1), Tail (0)
- Exam result → Pass (1), Fail (0)
- Light bulb → Working (1), Not working (0)

Main Types of Continuous Probability Distributions

A continuous distribution is used when data can take **infinite values within a range**

Uniform Distribution:

A **Uniform Distribution** is a type of continuous probability distribution in which **all values within a given range have equal probability** of occurring.

For example:

- Random number between 1 and 10
- Equal chance of selecting any value in an interval

Normal Distribution:

A **Normal Distribution** is the most important continuous probability distribution in statistics. It is a **bell-shaped symmetric curve** where most values are near the center (mean).

Height of people
Weight of students
IQ scores
Exam marks
Measurement errors

Statistical Problem Solving Process

The **Statistical Problem Solving Process** is a step-by-step method used to collect, analyze, and interpret data to solve problems and make decisions.

It is widely used in Data Science, business, research, and statistics.

Steps for Problem Solving Process



Interpret Results

Interpret Results is the step in the statistical problem-solving process where we **understand the meaning of analyzed data** and explain what the results show.

It helps us convert numbers into useful information for decision-making.

Meaning

After calculating values like:

- Mean
- Median
- Standard deviation
- Percentages
- Graphs

we study them carefully to find:

- Patterns
- Trends
- Relationships
- Conclusions

Example

Suppose the average marks of a class are:

Subject	Average Marks
Mathematics	42
Science	75
English	68

Interpretation:

Students are weaker in Mathematics compared to other subjects.

Purpose of Interpretation

- To understand what data tells us
- To identify problems
- To make better decisions
- To explain findings clearly